RIZ MUGHAL

QUIZ MASTER

Grand Quiz(CS502)

100% correct solution.

For more information you can visit my channel and for any type of help related to CS619 you can contact me.

YOUTUBE CHANNEL:

https://www.youtube.com/channel/UCINsFwDiB62SValCcPDZbRQ/playlists

FACEBOOK GROUP:

https://www.facebook.com/groups/923887914750307

CS502:Grand Quiz

Quiz Start Time: 10:21 AM, 04 January 2021

Question # 1 of 30 (Start time: 10:21:38 AM, 04 January 2021) **Total Marks:** 1

If we have an equation 8n2+7f*n+ 5f+6 then n is large, _____term will be much larger than the n term and will dominate the running time.

Select the correct option

- f(g(n)
- g(n)^2
- n^2
- f(n)

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Question # 1 of 30 (Start time: 10:21:38 AM, 04 January 2021) **Total Marks:** 1

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- ◉ n^2
- ○ f(n)

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Question # 2 of 30 (Start time: 10:22:04 AM, 04 January 2021)

Pseudo code of algorithms are to be read by _____.

Select the correct option

- (●) People
- ( ) RAM
- ( ) Computer
- ( ) Compiler

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Quiz Start Time: 10

Question # 3 of 30 (Start time: 10:22:23 AM, 04 January 2021)

In generating Fibonacci Sequence, we can avoid unnecessary repetitions by _____ process.

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Select the correct option

- ○ Tokenization
- ◉ Memoization
- ○ Randomization
- ○ Memorization

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Question # 4 of 30 (Start time: 10:22:40 AM, 04 January 2021)

For _____ values of n, any algorithm is fast enough.

Select the correct option

- (●) Small
- ( ) Medium
- ( ) Large
- ( ) Infinity

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Quiz Start Time:

Question # 5 of 30 (Start time: 10:22:56 AM, 04 January 2021)

Identify a TRUE statement about Knapsack.

Select the correct option

- ○ The Knapsack problem does not belong to the domain of optimization problems.
- ● The Knapsack problem belongs to the domain of optimization problems.
- ○ The Knapsack problem can not be solved by using dynamic programming.
- ○ The Knapsack problem is optimally solved by using brute force algorithm.

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Question # 6 of 30 (Start time: 10:23:16 AM, 04 January 2021)

What is the average running time of a quick sort algorithm?

Select the correct option

- ○ O(n^2)
- ○ O(n)
- ◉ O(n log n)
- ○ O(log n)

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Question # 7 of 30 (Start time: 10:23:35 AM, 04 January 20

If input "n" is odd, then median will be _____

Select the correct option

- (n+1)/2
- n/2
- (n-1)/2
- n + 2

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Quiz Start Time: 10:2

Question # 8 of 30 (Start time: 10:23:50 AM, 04 January 2021)

_____ provides us more accurate result, when input values are not closer with each other.

Select the correct option

- ○ Average
- ◉ Median
- ○ Mode
- ○ Mean

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Question # 9 of 30 (Start time: 10:24:05 AM, 04 January 2021)

The worst case running time of quick sort algorithm _____.

Select the correct option

- ○ Cannot be quadratic
- ◉ Is quadratic
- ○ Is always Exponential
- ○ Is linear

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Question # 10 of 30 (Start time: 10:24:22 AM, 04 January 2021)

Radix sort is a _____ integer sorting algorithm.

Select the correct option

- ○ Comparative
- ◉ Non-comparative
- ○ In-Place
- ○ Unstable

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Quiz Start Time: 10:2

Question # 11 of 30 (Start time: 10:24:36 AM, 04 January 2021)

_____ overcomes the limitations of _____ by working as per positional notations of numbers.

Select the correct option

- ○ Bubble sort, Radix sort
- ○ Counting sort, Radix sort
- ○ Radix sort, Bubble sort,
- ◉ Radix sort, Counting sort

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Quiz Start Time: 10:21

Question # 12 of 30 (Start time: 10:24:52 AM, 04 January 2021)

In merge sort algorithm, to merge two lists of size n/2 to a list of size n, takes _____ time.

Select the correct option

- (●) Theta (n)
- ( ) Theta log(n)
- ( ) Theta log2(n)
- ( ) Theta nlog(n)

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Quiz Start Time: 1

Question # 13 of 30 (Start time: 10:25:06 AM, 04 January 2021)

If the time complexity of an algorithm is O(n), then it is called _____ time complexity.

Select the correct option

- (●) Linear
- ( ) Constant
- ( ) Average
- ( ) Exponential

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Question # 14 of 30 (Start time: 10:25:21 AM, 04 January 2021)

The asymptotic growth of n(n+1)/2 is:

Select the correct option

- (●) $O(n^2)$
- ( ) O(n)
- ( ) O(n+2)
- ( ) O(n log n)

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Question # 15 of 30 (Start time: 10:25:37 AM, 04 January 2021)

We can make _____ recursive calls in Fibonacci Sequence.

Select the correct option

- (●) Infinite
- ( ) Finite
- ( ) Only one
- ( ) Zero

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Question # 16 of 30 (Start time: 10:25:52 AM, 04 January 2021)

Brute-force algorithm for 2D-Maxima is operated by comparing _____ pairs of points.

Select the correct option

- ○ Two
- ○ Some
- ○ Most
- ◉ All

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Question # 17 of 30 (Start time: 10:26:08 AM, 04 January 2021)

For quick sort algorithm, partitioning takes theta _____.

Select the correct option

- (n)
- log(n)
- nlog(n)
- n2log(n)

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Question # 18 of 30 (Start time: 10:26:24 AM, 04 January 2021)

In the following code the statement "cout<<i;" executes ---------- times.

```
for (int i = 1; i<=n ;i++)
   cout<<i;
```

Select the correct option

- (●) n times
- ( ) n+5 times
- ( ) Infinite times
- ( ) Zero times

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Quiz Start Time: 10:21 AM, 04 .

Question # 19 of 30 (Start time: 10:26:42 AM, 04 January 2021)

While Sorting, the ordered domain means for any two input elements x and y _____ satisfies only.

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Select the correct option

- ○ x < y
- ○ x > y
- ○ x = y
- ◉ All of the above

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Quiz Start Time: 10:21 AM, 04 January 20

Question # 20 of 30 (Start time: 10:27:00 AM, 04 January 2021)

Total Marks

In asymptotical analysis of n*(5 + 2) - 3 , as n becomes large, the dominant (fastest growing) term is some constant times _____

Select the correct option

- n_1
- n
- n+1
- n*n

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Quiz Start Time: 10:21 A

Question # 21 of 30 (Start time: 10:27:15 AM, 04 January 2021)

The main purpose of mathematical analysis is measuring the _____required by the algorithm.

Select the correct option

- ○ Space
- ○ Execution time
- ○ Inputs & outputs
- ◉ Execution time and memory

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Quiz Start Time: 10:21

Question # 22 of 30 (Start time: 10:27:32 AM, 04 January 2021)

In plane sweep approach of solving geometric problems, a _____is swept across the plane.

Select the correct option

- (●) Line
- ( ) Plane
- ( ) Cube
- ( ) Box

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Quiz Start Time: 10:21 AM, 04 January 2021

Question # 23 of 30 (Start time: 10:27:47 AM, 04 January 2021) **Total Marks:** 1

If we associate (x, y) integers pair to cars where x is the speed of the car and y is the negation of the price. High y value for a car means a _____ car.

Select the correct option

- ○ Fast
- ○ Slow
- ○ Expensive
- ◉ Cheap

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Question # 24 of 30 (Start time: 10:28:03 AM, 04 January 2021)

Which of the following is calculated with Big Omega notation?

Select the correct option

- ○ Medium bounds
- ○ Upper bounds
- ◉ Lower bounds
- ○ Both upper and lower bounds

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Quiz Start Time: 10:21 AM, 04 January 2021

Question # 25 of 30 (Start time: 10:28:19 AM, 04 January 2021)

Total Marks:

_____ is not useful measure of central tendency of given input set especially when the distribution of values is highly skewed.

Select the correct option

- Mean
- Mode
- Average
- Median

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Quiz Start Tir

Question # 26 of 30 (Start time: 10:28:36 AM, 04 January 2021)

Dynamic Programming approach is usually useful in solving _____ problems.

Select the correct option

- ○ normal
- ◉ optimization
- ○ array
- ○ loop

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Quiz Start Time: 10:2

Question # 27 of 30 (Start time: 10:28:52 AM, 04 January 2021)

One of the limitation in 0/1 knapsack is that an item can either be_____ in the bag or not.

Select the correct option

- ○ Use
- ◉ Put
- ○ Move
- ○ Store

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Question # 28 of 30 (Start time: 10:29:06 AM, 04 January 2021)

In Heap Sort algorithm, the total running time for Heapify procedure is _____.

Select the correct option

- ○ Theta (log n)
- ◉ Big-oh(log n)
- ○ Omega (log n)
- ○ O (1) i.e. Constant time

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Quiz Start Time: 10:21 A

Question # 29 of 30 (Start time: 10:29:22 AM, 04 January 2021)

An algorithm is said to be correct if for every _____ instance, it halts with the correct _____.

Select the correct option

- (●) Input , Output
- ( ) Design , Analysis
- ( ) Design , Analysis
- ( ) Key , Analysis

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Quiz Start Time:

Question # 30 of 30 (Start time: 10:29:39 AM, 04 January 2021)

In Heap Sort algorithm, the maximum levels an element can move upward is _____.

Select the correct option

- (●) Theta (log n)
- ( ) Big-oh(log n)
- ( ) Omega (log n)
- ( ) O (1) i.e. Constant time

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